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Dynamic capabilities for co-innovation in timber construction: a case study of Swedish cross laminated timber innovation ecosystems

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Introduction: Despite growing recognition of engineered wood products as a critical enabler of low-carbon and circular construction, the diffusion of cross laminated timber (CLT) remains comparatively slow relative to its technical maturity and sustainability potential. This suggests that timber adoption is shaped not only as a design challenge but also by firms' capabilities, collaborative practices, and the broader conditions under which innovations are developed and commercialized. Such conditions include organize renewal, manage interdependencies, allocate responsibilities, and align around shared value propositions across projects. This study explores how dynamic capabilities, understood as situated practices embedded in recurring cross-project interactions, may enable co-innovation in CLT-based new-build and renovation across projects within the contextual boundary of emerging innovation ecosystems (IE).
Methods: Drawing on an exploratory case study of Swedish CLT-based construction, the empirical material consists of transcribed interviews, focus groups and observation notes representing and contextualizing activities indicative of dynamic capabilities among collaborating firms. Thematic analysis was applied for developing activities into capabilities.
Results: Activities were developed into three levels: project-specific value proposition reconfiguration and resource coordination (Level 1, operational capabilities), joint development and technology implementation (Level 2, dynamic capabilities), and knowledge exchange and actor alignment (Level 3, dynamic capabilities). Capability development occurred through iterative adaptations, informal coordination, and relational negotiations rather than fixed or rule based processes, making it emergent and context-dependent.
Discussion: These findings show that IE structures may create conditions that facilitate collaborative adaptation and learning, potentially supporting innovation beyond individual projects. By foregrounding co-innovation at the firm level, the study shows how these dynamic capabilities can create conditions for scaling CLT adoption beyond isolated projects, supporting broader timber uptake and sustainability ambitions.

KEYWORDS
co-innovation, construction management, cross laminated timber, dynamic capabilities, innovation ecosystem, networks, timber construction

1 Introduction

Innovations in engineered wood products (EWPs) have expanded the structural capabilities of timber, enabling applications in mid- and high-rise construction and prompting interest in wood as a renewable, low-carbon alternative to conventional materials (Himes and Busby, 2020; Ilgin et al., 2023b). Particularly, cross laminated timber (CLT) is positioned as a material with substantial potential to replace carbon-intensive alternatives in urban development (United Nations and the Food and Agriculture Organization of the United Nations, 2021; Ilgin et al., 2023a), both in new housing production but also for renovation and extension of existing buildings, encouraged by the comparatively low weight of timber in relation to its structural properties (Holtström et al., 2024). Despite this progress, the potential of EWPs to contribute to climate objectives remains largely unrealized. Even in mature market contexts such as Sweden, where approximately 90 percent of single-family homes and around 16 percent of multi-story buildings are constructed using prefabricated timber construction (Trähusbarometern, 2025), innovation has not scaled at the pace required to shift the broader material landscape of construction (Erikshammar et al., 2024). While prefabrication and vertically integrated housebuilding have supported timber uptake, EWPs such as LVL or CLT have not yet achieved comparable market maturity, underscoring the need for coordinated firm-level development and innovations.

Transitioning construction to sustainable practices requires transformation of business models and value networks (Rao et al., 2025). Scaling timber is therefore not only a building design challenge, but also a capability challenge for firms. The limited scaling of EWP:s highlights a fundamental point: while innovations are conceived, developed, and operationalized within firms, products as CLT are dependent on co-innovation for successful integration with complementary materials (Ilgın and Karjalainen, 2024). Hence, the capabilities for technological advancement should be examined at the firm level, where strategic priorities and collaborative arrangements are operationalized. From this perspective, the capabilities and innovation practices of individual firms, operating within broader ecosystems (Jacobides et al., 2024), influence the pace of timber-based innovation and application, in line with Kapoor and Lee's (2013) notion on the importance of development of complementarities for the success of new technologies.

Fragmentation has been shown to be associated with reduced learning opportunities and limited cross-project continuity, which may constrain the diffusion of new practices and technologies (Hughes and Stehn, 2019). Fragmentation refers to the separation of roles, firms, and project phases across procurement, governance, and supply, where short-term, contract-based constellations hinder innovation development and learning across projects. Several approaches have been explored to address these challenges: studies have examined the influence of practice-based inputs at the project, firm, and industry levels (Ozorhon and Oral, 2017), investigated how to overcome organizational lock-in (Eriksson and Szentes, 2017), and analyzed procurement-driven initiatives (Larsson et al., 2022). While these works have advanced understanding of how innovation emerges within individual projects, they also highlight a persistent challenge: project-level innovations rarely scale to construction industry-wide transformation (Stehn and Erikshammar, 2025) pointing to a developed form for innovation governance. This limitation has important implications for the transition to low-carbon construction.

Recent work suggests that co-innovation may offer a pathway for accelerating the development and diffusion of EWPs by fostering shared learning, integration of financial, human and knowledge resources, and coordinated activities across interdependent actors (Ozorhon and Oral, 2017). In timber construction, this is increasingly important as emerging industrial investments and sustainability ambitions demand innovation capabilities that extend beyond single projects and require alignment across networks (Erikshammar et al., 2024). This is also discussed more broadly in relation to the importance of collaboration for sustainable innovation in prefabricated construction firms (Guo and Li, 2022). Understanding co-innovation as an enacted, practice-based process aligns with the dynamic capabilities (DC) framework, which conceptualizes how organizations sense and seize opportunities and reconfigure resources in changing environments (Teece, 2007). DCs are interconnected with operational capabilities, that are included in the capability framework in this study, adapted from Collis (1994). Adopting this view, and following Romero and Molina (2011), co-innovation is understood here as a process of resource and risk sharing enacted through informal and emergent practices, i.e., a dynamic process explaining how co-innovation unfolds and how it may enable CLT scaling beyond isolated projects. It also suggests that these enacted activities can be interpreted through the DC framework proposed by Teece (2007). In this article, situated practices are central to understanding how DCs are enacted through everyday activities rather than as abstract transferable assets (Collis and Anand, 2021). This interpretation addresses critiques of the DC theory's elusiveness in construction (Green et al., 2008) and aligns with calls for contextualized, practice-oriented approaches (Schilke et al., 2018).

In other industries, co-innovation has been shown to emerge within innovation ecosystems (IEs) as networks of actors working together toward a focal value proposition (VP) (Adner and Kapoor, 2010; Adner, 2016; Marcon and Ribeiro, 2021; Stonig et al., 2022). In construction, IEs are described as multilateral networks where actors interact beyond contractual ties, creating conditions for scaling innovation (Deep et al., 2021; Erikshammar et al., 2024). This is similar to business ecosystems, which have been shown to increase sustainability through prefabrication in individual timber-based residential buildings (Toppinen et al., 2022). To explore how collaborative dynamics shape co-innovation across projects, IEs are introduced as the system boundary for identifying interactions and tracing how DCs materialize across projects. The IE boundary clarifies ecosystem-level co-innovation interactions versus isolated project efforts and frames how DCs materialize over time.

Although timber has become central to discussions of sustainable building design and construction, little empirical research has examined how co-innovation shapes the development and scaling of EWPs across projects. To address this gap, this article investigates the following research question:

How do dynamic capabilities support CLT-based co-innovation within and across building projects?

By focusing on the interplay between DCs, CLT-based co-innovation practices, and IE-level coordination, the study aims to contribute new insights into the capabilities required to accelerate the scaling of EWPs and support the sustainability ambitions of the built environment, including both new-build and renovation. This is done by a case study that investigates co-innovation activities undertaken by firms collaborating primarily in building new-build CLT-projects, reflecting the CLT-based construction activities within the case region. Activities relating to the extension projects that have been executed by the networking firms have not been explicitly investigated in this study; the results and discussion however include capabilities drawn from both new-build and renovation. The analysis is drawn on an aggregated capability level that is expected to be applicable to both practices.

2 Dynamic capabilities for co-innovation in construction

This section establishes the theoretical foundation by defining the key constructs: dynamic capabilities understood as enacted and situated activities; co-innovation as joint development across projects; and the IE as the multilateral alignment around a focal value proposition. These constructs are analytically distinguished but empirically intertwined in construction contexts shaped by fragmentation and project-based delivery.

2.1 Definition of dynamic capabilities

The construction sector's fragmented structure and project-based logic pose persistent challenges to the development and scaling of innovation, particularly in the context of advancing EWP for sustainable building design. Under such conditions, DCs offer a useful lens for understanding how co-innovation emerges through situated, interactional practices that connect actors across projects. Introducing the IE as an analytical boundary makes it possible to distinguish between project-level activities and broader systemic coordination mechanisms, thereby enabling a more nuanced examination of how DCs are enacted beyond individual projects.

In this study, DCs are understood as firms' abilities to integrate, build, and reconfigure resources in response to changing environments (Teece et al., 1997). In DC terminology, Teece (2007) frames activities as microfoundations that collectively constitute an organization's ability to adapt. However, the position taken herein views DC as situated practices enacted through routinized everyday activities (Eisenhardt and Martin, 2000) embedded in project-based co-innovation interactions. This perspective allows these microfoundations to be interpreted not as separate from practice but as constitutive elements of it. Recent work has examined DCs at both the industry level and within specific construction applications (Li et al., 2025; Robson et al., 2024) and has considered how DCs are enacted by loosely connected actors across temporal (project) and permanent organizations (Adam and Lindahl, 2017). Li et al. (2025) identified positive reinforcement between construction start-ups' platforms and DCs, suggesting that construction firms can potentially scale innovative business models across projects by combining platform collaboration and DCs. Najafi-Tavani et al. (2018) discuss the interplay between innovation and absorptive capabilities and co-innovation networks, suggesting that both capabilities are vital for the performance of co-innovated products. In this article, DCs are understood as enacted activities that enable co-innovation across projects by:

- Sensing opportunities for joint development,
- Seizing these opportunities through collaborative routines and informal coordination, and
- Transforming organizational arrangements to sustain innovation beyond single projects.

This article conceptualizes IEs as multilateral networks that define the exploratory boundary for studying co-innovation, where the IE can be viewed as a shared organization that contains DCs for co-innovation (Bossink, 2002). Following (Adners (2016), p.42) definition of IEs as "the alignment structure of the multilateral set of partners that need to interact in order for a focal value proposition to materialize", the IE perspective emphasizes the systemic nature of the focal value proposition (VP) (Kapoor, 2018; Foss et al., 2023). The realization and scaling of such a VP depend on interconnected actors whose interdependencies, extending beyond contractual ties, can support scaling of innovation (Deep et al., 2021).

Delineating the IE boundary focuses the analytical attention on capability-building beyond single building projects and how this supports collaborating firms in creating and capturing value from co-innovation, thereby motivating the scaling of innovations in CLT-based construction. From an IE perspective, value capture extends beyond transactional profit to include learning, innovation capacity, and trust (Ben Letaifa, 2014). Accordingly, value capture is understood as a dynamic process through which actors, collectively and individually, benefit from co-innovation through efficiency gains, increased sales, and resource sharing. When properly managed, the total value captured from the value proposition of the IE is larger than the sum of all complementarities (Baldwin et al., 2024). These characteristics make the IE perspective suitable for examining how DCs evolve in the co-creation of a VP (Adner, 2016; Adner and Kapoor, 2010) within and across projects.

2.2 Analytical framing: a capability-based model for studying co-innovation

While DC theory offers a promising approach to understanding adaptation for co-innovation, concerns have been raised about its direct applicability to construction (e.g., Green et al., 2008; Adam and Lindahl, 2017). In particular, Green et al. (2008) suggest DCs to be analyzed as practices enacted by individuals rather than as corporate strategy. A contextualized approach is therefore essential (Schilke et al., 2018; Robson et al., 2024). Therefore, the position herein is that DCs refer to situated activities through which organizations sense, adapt, and reconfigure across projects. These activities are treated as situated because they arise from the specific project constellations, inter-firm dependencies, and ecosystem

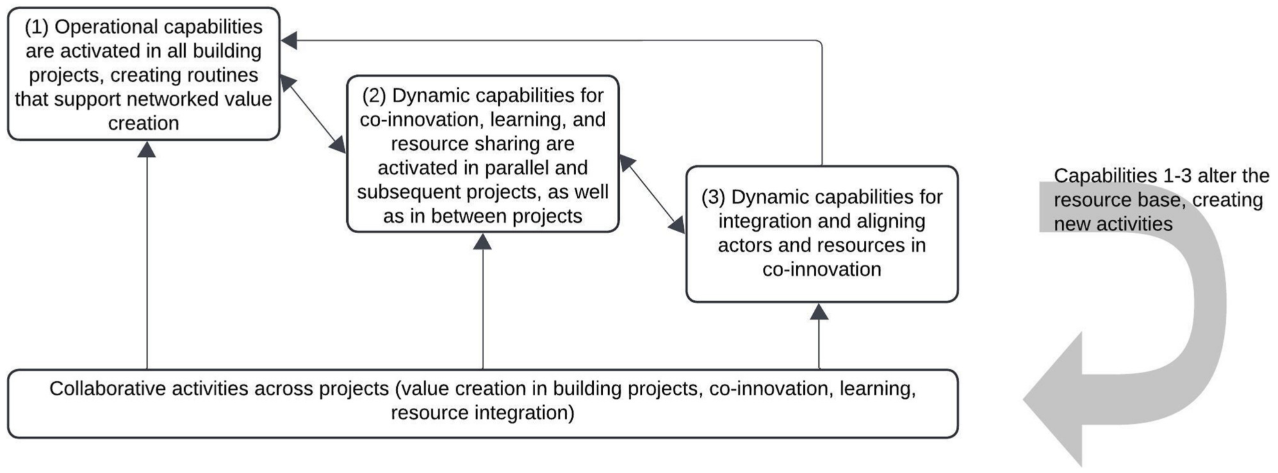

FIGURE 1
The three capability levels as concurrent and interdependent layers shaped by activities across projects. Arrows indicate influence or mutual influence (authors' own creation).

interactions that shape how capabilities materialize across projects. Drawing on Collis (1994) original four-level capability framework, this study adopts a three-level model that excludes higher-order DCs while still acknowledging that learning processes generate routines that shape capability development as originally considered in Collis (1994). In a construction context, this progression spans project-level operational routines, through learning, to co-innovation and strategic development. Co-innovation denotes joint efforts that combine resources and learning beyond a single project to refine and implement solutions. The analytical framing conceptualizes how observable activities may reflect underlying capabilities, and how these interact with IE structures, as co-innovation emerges through networked interactions across three levels.

- The first level of Collis' framework concerns operational capabilities, i.e., the routines and enacted activities (Winter, 2003) that enable consistent value creation in building projects (Harty, 2005). These capabilities are typically embedded in project execution and resource coordination and are shaped by industry norms and contractual arrangements (Winch and Leiringer, 2016). Within an IE perspective, operational capabilities support the realization of a focal VP by aligning individual offerings (Adner, 2016) in project settings, led by the focal actor as a systems integrator (Rutten et al., 2009; Lavikka et al., 2021).
- The second level captures DCs that enable the refinement and adaptation of operational routines across projects (Teece et al., 1997). In construction, projects are temporary constellations embedded within permanent organizations (Dubois and Gadde, 2002), making collaboration central to innovation. Using the IE as an analytical perspective reveals how actors leverage network relations to extend individual VPs and co-develop new solutions beyond individual projects (Erikshammar et al., 2024). Alignment mechanisms such as cross-project learning and resource sharing support adaptation of routines across projects.
- The third level DCs support the orchestration of resources and actors involved in shaping the focal VP. These relational capabilities often involve alliance-based learning and coordination across organizational boundaries (Helfat and Raubitschek, 2018; Qiu et al., 2022). Learning processes at this level generate new routines that evolve lower-level capabilities (Zollo and Winter, 2002; Schilke, 2014).

In the analytical framing (Figure 1), the three capability levels should be understood as concurrent and interdependent layers of activity. Operational capabilities are continuously enacted within projects, while level 2 DCs emerge when actors adapt to these routines across projects, and level 3 DCs appear when alignment is required across the broader actor constellation. These layers interact over time as recurrent project linkages enable learning and adaptation, but no fixed temporal sequence governs their development. Rather, the levels represent different analytical expressions of capability enactment within the evolving IE context.

The IE is treated as the alignment structure around a focal VP, rather than a formally centralized entity; this distinction is central in project-based construction where fragmentation persists across phases, actors, and projects. An IE analysis therefore highlights how leaders may use DCs to facilitate alignment to support and benefit from shared VPs (Foss et al., 2023). Alignment mechanisms include trust-building and joint governance structures that stabilize interdependence among actors. Integrative DCs (Helfat and Raubitschek, 2018) such as identifying opportunities, aligning interests, and coordinating efforts play a key role in enabling co-innovation.

3 Materials and methods

The study adopts an exploratory case study design (Eisenhardt, 1989; Dubois and Gadde, 2002a) to investigate how DCs, understood as situated practices enacted through routinized activities, enable co-innovation within the contextual boundary of an IE. Further, a single-case approach is chosen as advocated by Ambrosini and Bowman (2009) for being able to analyze firm-specific DCs. DCs are not directly observable constructs; rather, they are manifested through patterns of action, interaction, and decision-making over time. Capturing such patterns requires close engagement with actors' behaviors, and adjustments within the studied context.

Co-innovation within an IE involves multiple interdependent actors, overlapping roles, evolving governance structures, and continuous alignment around a focal VP. Under such changing conditions, the boundaries between a phenomenon and context are inherently blurred. The development and enactment of DCs are embedded in relational dynamics and cannot be meaningfully separated from the ecosystem in which they unfold. In such situations, a qualitative approach is appropriate because it enables investigation of evolving practices that cannot be isolated or experimentally controlled (Eisenhardt and Graebner, 2007). The research design supports processual analysis and allows theoretical elaboration grounded in rich empirical material.

IEs are approached as empirical networks that define the system boundary for observing how actors align around a focal VP and how these alignments condition capability development. In defining the IE boundary, actors are included whose co-innovation activities directly contribute to the focal VP for CLT-based structural framings across multiple projects. Actors involved solely through single-project, transactional relationships are outside the analytical scope, unless their activities demonstrably shape cross-project learning or coordination around the VP. Accordingly, the study does not claim to "observe DCs directly". Instead, it examines activities and interactions that indicate how capabilities materialize in the co-innovation context. Through this design, the study contributes theoretically by unpacking the microfoundations of DCs in an IE setting rather than by estimating predefined quantitative effects.

3.1 Case and sample selection

The empirical setting is timber-based construction in mid-Sweden, where collaboration between industry, academia, government, and civil society has shaped a regional IE (Carayannis et al., 2018). This context combines mature industrial capacity, policy development, and large-scale investments, creating favorable conditions for studying how organizations sense opportunities, reconfigure resources, and renew collaborative arrangements for co-innovation. Firms in this setting have long engaged in product development and cross-organizational collaboration, making it a suitable environment for examining how DCs evolve in practice. This region hence provides a favorable context for IE and follows a critical case strategy to examine microfoundations of DCs in accordance with Flyvbjerg (2006).

Theoretical sampling guided actor selection to illuminate relationships among key constructs (Eisenhardt and Graebner, 2007). Sampling dimensions included: (a) CLT new-build and renovation projects utilizing novel technology with potential for further innovation; (b) resourceful actors capable of investing in product and process development; (c) actors engaged in product or process innovations originating from co-innovation activities, across stages of the innovation process (ideation, development, testing, commercialization); and (d) complementary actors providing architectural customization, supporting components (e.g., fasteners, membranes), and integrated systems (e.g., insulation, windows).

Several co-innovation activities related to CLT were evident, including development and testing of preassembly of moisture membranes in the CLT-factory (product and process innovation), testing and evaluating assembly support software (process and service innovation), and development of and investment in production for prefabricated floor slabs combining CLT and glue laminated timber (product innovation).

Sampling was grounded in field observations and six initial interviews used to identify all larger CLT-based projects (>100 m^3) executed in the region during 2021–2022, yielding a total of 15 projects. Subsequently, 27 short telephone interviews (15–30 min) were conducted to map supply chain actors involved in CLT framing for these projects, as well as associated architects, technical consultants, and clients. Informants represented diverse project roles, including construction management consultants, contractors, technical consultants, customers, and engineered wood product producers. Information was iteratively validated through follow-up interviews, and in some cases the same informant was re-interviewed to corroborate or clarify data.

The actor (c) selection is based on the identified project relationships (b, d) and observed co-innovation activities. To examine capabilities for co-innovation, we deliberately selected actors actively engaged in such activities, consistent with critical case logic. This expected selection bias among respondents was thus considered supportive for the study. The actor sampling was further complemented by actors and customers not directly involved in the observed co-innovation activities to provide contrasting perspectives.

All selected actors agreed to participate. Interviewees represented both strategic and operational perspectives and were directly involved in CLT-related innovation activities and/or project execution. Relationships among actors encompassed both contract-based building project deliveries and non-contractual co-innovation activities conducted within and across projects. These relationship types were mutually reinforcing. In some instances, supplier-initiated co-innovation led to subsequent project-based contracts for testing new solutions. In others, long-standing project collaborations evolved into co-innovation initiatives, supported by accumulated trust and shared objectives.

Figure 2 illustrates how the focal VP is commercialized through the contractual roles in building project deliveries following Adner and Kapoor (2010). The development of the focal VP through co-innovations was not contract-bound but emerged through multilaterally networked interactions (not shown in Figure 2) around the focal VP for CLT-based structural framings. Within this IE boundary, framing contractors acted as focal actors, integrating and commercializing suppliers' individual VPs into a system-level focal VP.

3.2 Data collection and analysis

Data collection and analysis proceeded in three phases:

1. Identification and validation of contextualized practices
2. Data collection through interviews and public seminars
3. Thematic analysis of potential DCs

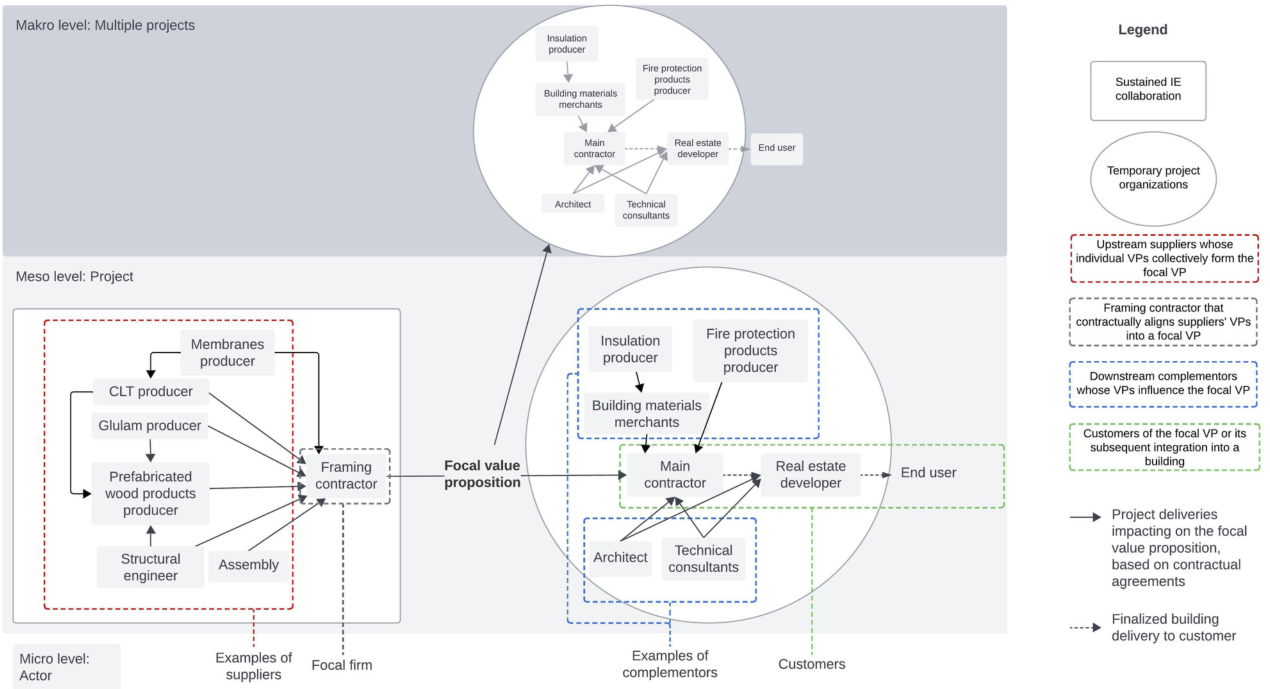

FIGURE 2
Building project contractual positions of roles (authors' own creation).

3.2.1 Identification and validation of contextualized practices

A literature review established the theoretical foundation of DC and IE research and identified studies applying these frameworks in construction. This review informed the development of 19 tentative DCs (Table 1) derived from Teece (2007), IE literature (Adner and Kapoor, 2010; Adner, 2016), and a sector-specific study (Linde et al., 2021). Following Schilke et al. (2018), these tentative DCs were reformulated into recognizable, context-specific activities informed by industry knowledge. To establish content validity, that is, the extent to which these empirical indicators (contextualized practices) accurately capture DCs, is a methodological challenge (e.g., Nielsen, 2014). We followed the recommendations of Hardesty and Bearden (2004) and conducted focus group discussions with 67 participants, organized into 11 groups, across the regional timber-based construction value chain. The 19 DC-related activities were distributed among the groups and discussed, with each group reviewing three of the activities and discussing their meanings, examples, and challenges for adoption within timber-based building projects. This ensured that the indicators (contextualized DC-related activities) were grounded in practitioners' interpretations rather than abstract theory, providing expert content validation of the indicators. The research team then formulated seven contextualized interview questions (Table 1).

3.2.2 Data collection

The first author conducted all data collection and transcriptions, with a co-author continuously reviewing procedures to ensure quality and validity. All interviews were based upon the same set of interview questions (Table 1), to ensure data triangulation by obtaining responses from actors holding different roles within the IEs. Even though questions were outlined around the individual actors, they also aimed at investigating the networked activities and relationships, which provided us with data on the same activities from different actor perspectives. The interviews were initiated around collaborations in one or several of the 15 previously identified CLT-based building projects. Respondents were frequently reflecting on earlier, concurrent and planned projects, thereby providing insights that extend beyond a snapshot of any individual project.

The data, consisting of audio recordings and transcripts were stored in a research database accessible only to the research team. Transcript files were imported into NVivo 14 for thematic analysis. Data collection (Table 2) was conducted between December 2022 and July 2024 and included seven semi-structured interviews, two semi-structured group interviews, and observations from two public seminars that presented insights from a co-innovation project. Informants were operations managers or highly specialized professionals actively engaged in co-innovation and holding a mandate to influence its development. Interviews typically lasted 40–45 min (one shorter session of 15 min) and the structure in Table 1 to elicit reflections on embedded activities and interactions across projects.

3.2.3 Thematic analysis

Thematic analysis followed Braun and Clarke (2006) six-step process, adopting reflexive principles (Braun and Clarke, 2021) to let researchers' professional experience form interpretation. An audit trail was maintained, documenting the coding steps.

Initially, transcripts were coded by the first author into 25 potential, context-dependent, operational activities (Table 3),

TABLE 1 Contextualization of dynamic capabilities for development of interview questions.

21 tentative dynamic capabilities	Contextualized microfoundations for focus groups	Overarching interview questions
Capability for developing the value proposition[a]	We continuously adapt and improve our offering to address the customers' current and future needs	How do you present the VP to customers?
Capability for shared business model development with incentives for all partners	We conduct business that generates benefits for all partners in the collaboration	
Capability for reconfiguring the offering to customer needs	We systematically reevaluate and adapt the shared customer offering on a project-by-project basis	
Sensing capability represented in top management	Our management monitors and interprets market and technological developments in the external environment	How, and by whom, is business intelligence managed related to the VP?
Capability for sensing new business opportunities	Our management allocates resources for the for search and identification of new business opportunities	
Capability for exploring opportunities to new solutions	Our management allocates resources for research and innovation aimed at developing new solutions	How is R&D managed related to the VP and with whom do you collaborate in this?
Capability for finding and establishing co-innovation partnerships	We participate in research and innovation projects together with partners beyond our own organization	
Capability for co-innovation	We engage in structured co-innovation processes with our partners to jointly advance innovation and business development	
Capability for learning	We have routines for knowledge transfer with partners and customers, capturing lessons and opportunities after each completed building project	How is the transfer of knowledge between projects managed, internally and with partners?
Capability for searching new resources	We regularly meet and engage with potential partners to identify and access new resources for business and development	How is the collaboration within the region developed and how do you find new partners?
Capability for structural leadership of the network	We maintain clear and jointly agreed-upon role allocations within the collaboration surrounding the customer offering	
Capability for building loyalty and commitment in the network	We actively work to strengthen and further develop the team collaborating on the customer offering	
Capability for integration of external specialized resources	We integrate the products and services of external organizations into our customer offering	
Capability for development of contract-based incentives	We design and refine contract models to ensure mutually beneficial incentives for all partners	
Capability for partner alignment	We dedicate resources to continuously align and strengthen collaboration with partners across projects	
Capability for seizing investments for long-term development	Our management makes strategic investment decisions that support the long-term development of the business	How do you develop the business around the VP to meet current and future customer needs, and what investments are taken by you or your partners?
Capability for seizing investments for new businesses	Ou management invests in initiatives that open new business areas	
Capability for mitigating investment risk within the network	All partners jointly invest in the shared business to mitigate risks	
Capability for decentralized decision-making[a]	Decisions concerning the development of the shared business are made close to the customer	
Reconfiguring capability represented in top management	Our management proactively challenges and reevaluates existing organizational structures	How do you make sure you keep out of established patterns in business and projects?
Capability for continuous renewal	We challenge 'the way we have always done things' and avoid lock-in effects from past investments	

[a]not validated by focus groups.

that actors described as part of co-innovation. Statements highlighting how these practices and activities contributed to value capture (e.g., efficiency gains, scaling benefits) were included, while routine administrative tasks were excluded as 'business as usual.' Activities reflecting cross-project learning and collaborative adaptation were prioritized as indicative of DCs.

As a next step, all three authors reviewed the codes, followed by the first author developing these into tentative themes. Subsequently, the tentative themes were discussed by all authors and collaboratively developed into a final set of 8 sub-themes (condensed activity patterns) and 5 themes (interpreted capabilities). This approach ensured researcher triangulation of

TABLE 2 **Overview of data collection events and participants.**

Actor/s and roles	Data collection activity
Framing contractor (FC1)	Interview
Framing contractor (FC2)	Interview
CLT producer (CLT1)	Interview
CLT producer (CLT2)	Interview
Prefabricated wood products producer (PW)	Interview
Main contractor (MC1)	Focus group interview
Architect (A1)	Interview
Architect (A2)	Interview
Real estate developer (RE1)	Interview
Fire protection solutions producer (FP)	Focus group interview
Insulation producer (IS)	Interview
FC1	Seminar speech
CLT2	Seminar speech
Membrane producer (M)	Seminar speech
Test institute (TI)	Seminar speech
A1, FC1, CLT1, CLT2, MC2, TI, Technical consultants (TC1, TC2), M, Business promoting organization (BPO)	Seminar group discussion
FC1, CLT2, M, Structural engineer (SE), RE2	Focus group discussion

the analysis. The sub-theme coding condensed the explored narratives mobilized by practitioners of how they enact and interpret co-innovation within the IE boundaries. Using an IE boundary perspective helped identify co-innovation activities e.g., VP reconfiguration, joint development, incentives, knowledge exchange, per level 1–3 (Collis, 1994):

1. Configuring the focal VP for project-specific solutions and coordinating resources to meet shared project-specific objectives.
2. Engaging in joint development efforts, implementing new technologies, and creating incentive structures that support innovation implementation.
3. Facilitating knowledge exchange, aligning actor roles, and managing structures to enable co-innovation.

The last step of the coding then grouped the 5 themes into three levels (Table 3, column #4) of DCs for co-innovation in CLT networks inspired by Collis (1994) and the IE structure. To ensure theoretical alignment, the coding condensed observed activities into sub-themes and themes that map onto the three capability levels used in the analysis. This provides a linkage from analytical framing to the empirical patterns presented in Section 4.

4 Results

The following section presents the empirical findings from the exploratory case study organized to illustrate how the defined capabilities are expressed through actors' situated practices within the specified IE boundary The analysis focuses on how DCs were observed through collaborative activities and actor interactions across timber building projects. By examining these processes, the findings illustrate how co-innovation was enacted in the network and associated with new or refined solutions. These findings are also framed by characteristics for IE governance in comparison with traditional governance structures in on-site construction.

4.1 Operational capabilities for value creation in CLT-based building projects

4.1.1 Capabilities for reconfiguring the value proposition to individual building projects

The VP was configured in the co-innovation process to allow project-specific reconfigurations, which were led by the focal firm and carried out by the project actors. Reconfigurations were typically triggered by client procurement strategies, technical conditions, and the focal firm's ambitions to improve efficiency and profitability. Typically, the focal firm would offer a fixed price for the CLT structure (the focal VP), relying on pre-negotiated fixed unit prices from the suppliers. When the contract with the main contractor was signed, the focal firm initiated a detailed design process of the structure together with the suppliers, with the aim of improving project economics within contractual constraints. These adjustments were reported to create new business opportunities for some actors by incorporating new products and processes:

TABLE 3 **Results from thematic analysis, aggregated to the three-level capabilities framework.**

Codes (operational activities)	Sub-themes (condensed activity patterns)	Themes (capabilities)	Aggregated dimensions for DCs in CLT networks
Optimizing resources through combining individual offerings	A VP that allows building project reconfiguration	Capabilities for reconfiguring the VP to individual building projects	**Operational capabilities (Level 1)** Value creation and value capture (Collis, 1994; Winter, 2003; Ben Letaifa, 2014)
Renegotiating the VP in the building project's procurement phase			
Reconfiguring the VP for the building project			
Developing financial incentives	Added value for actors through consistent value creation relations	Capabilities for exploiting specialized resources	**Dynamic capabilities (Level 2)** Development of operational capabilities for value creation (Collis, 1994; Teece et al., 1997; Linde et al., 2021; Qiu et al., 2022)
Achieving efficient business processes through repetitive project collaboration			
Extending the individual VPs			
Using the network for non-exclusive product development	Leveraging the network for co-innovation		
Performing research and development between projects			
Developing and implementing pre-fabrication			
Investing for VP			
Executing co-innovation activities			
Learning across organizations	Learning about the VP and its value creation	Capabilities for learning	**Dynamic capabilities (Level 3)** Alliance-based learning (Zollo and Winter, 2002; Teece et al., 1997; Qiu et al., 2022), for development of operational capabilities (Collis, 1994)
Individual actor learning			
Dissemination of knowledge			
Managing and sharing business intelligence	Building an operational network structure	Capabilities for actor alignment	**Dynamic capabilities (Level 3)** Strategic integration of internal and external resources (Collis, 1994; Teece, 2007; Foss et al., 2024), and alliance management (Helfat and Raubitschek, 2018; Schilke, 2014)
Aligning actors' positions			
Allowing decentralized decision-making			
Partnership scouting			
Mitigating risk within the new VP	Project risk driving and dictating new VPs		
Managing risk through outsourcing			
Building trust	Expanding relations	Leadership capabilities	
Aligning complementors' roles and relations			
Aligning partners' roles and relations			
Engaging partners through pioneering actions	Leading the shared vision		
Communicating the VP			

> '. . ./What are we trying to market? The opportunity to not only buy a CLT-frame, but a framing that is a bit smarter than that. To say that we can offer the use of different types of, particularly external walls, but possibly also load-bearing interior walls.' (PW1)

However, reconfiguration also reduced the delivery scope for other actors, raising concerns about whether the financial incentives were sufficient to justify their participation in optimization efforts:

> '. . ./Well, if you quote a framing and we then optimize it, reducing the volume of timber in the frame, that [volume] is then reduced from the contract with the supplier.' (FC1)

Some actors experienced a reduced business scope as the network evolved, yet these reconfigurations were regarded as necessary by several respondents to secure new projects and to remain competitive. Renegotiations often took place under time pressure during the procurement phase and depended on the design expertise within each organization to adapt and further develop the shared VP.

The observed activities were firmly grounded in traditional building project roles and operational activities. However, recurring collaborations and co-innovation efforts among actors were associated with a more integrated approach to project delivery. Through these continued interactions, the network reported additional value reflected in improved efficiency, new business opportunities, and the development of higher-value products.

4.1.2 Capabilities for exploiting specialized resources

Participation in the network was described as fostering stable and consistent business relationships, which respondents associated with improved efficiency through scaling effects, shared resources, and streamlined processes. Standardized routines, like shared design solutions and project management processes, were developed to reduce resource use compared to projects delivered outside the collaboration. The focal actors benefited directly from volume-based pricing, while the shared practices were said to contribute to lower project management costs across the network:

> '. . ./delivery worth 5 or 6 MSEK, in which I had approx. 8 hours of administrative work . . . while my colleague /. . ./ had a project worth 150 kSEK. She put approx. 80 hours in one month into a small project with a customer that had no knowledge of the process' [of designing and building in CLT]. (CLT2)

Actors using fixed pricing for predetermined deliveries reported higher margins from the shared routines. However, service providers relying on variable pricing did not achieve the same economies of scale from the new practices.

Actors collaborated on development initiatives through shared resources across organizational boundaries and over multiple building projects. Such resources included primarily competence held by individuals or organizations for developing innovations, and production facilities and building projects for testing them. One salient example cited by respondents is the integration of a moisture membrane into the industrial CLT production process. Three actors (the membrane producer, the CLT producer, and the framing contractor) said that they benefited from the implemented innovation (i.e., increased prefabrication):

> '. . ./One of the keys to making this happen was first and foremost to attach the membranes in the CLT factory. Since this was new to the CLT producer, we processed a solution through discussion [among all three partners]. The membrane supplier customized their product's widths both for the CLT factory and to our building site.' (FC1)

This innovation was not exclusive to the focal VP or the actors' value capture, as the involved actors would market the innovation outside the IE. Suppliers and complementors also targeted a broader market by developing technical guides for timber-based construction, created in collaboration but centered on individual products to support their businesses beyond the focal VP. The collaborative activities were reported to enable controlled full-scale testing and verification of technical innovations. Sustained partnerships appeared to provide conditions that enabled this work to continue despite challenging building projects conditions, where lowest-bid procurement and risk aversion typically constrain innovation.

Co-innovation occurred primarily between the focal firm and its suppliers, while the two product-providing complementors remained largely outside of these processes. A shared innovation platform did not fully materialize, as both complementors struggled to gain access to joint development activities. One complementor was further restricted by contractual obligations to an external material retailer, limiting its incentives to collaborate on specialized products aligned with the shared value proposition.

4.2 Dynamic capabilities for alliance-based learning and strategic integration of resources

4.2.1 Capabilities for learning

Opportunities for external learning were reported to emerge through active outreach and knowledge-sharing efforts. Both formal and informal channels were used, as actors sought new insights from external sources while disseminating knowledge generated through the network's co-innovation activities.

Collaboration initiatives for broader knowledge dissemination were carried out with support from regional business promotion organizations. These activities included events to which potential customers, partners, and other industry actors, including competitors, were invited:

> '. . ./One of our major goals has been to spread knowledge about new building practices to as widely as possible. We benefited greatly from [BPO2], it has helped us to reach out.' (SE)

Despite these deliberate dissemination efforts, the actors described limited routines for capturing and transferring lessons learned between projects, with knowledge transfer remaining largely reliant on individuals, reflecting established practices in the construction sector.

4.2.2 Capabilities for actor alignment

Product-supplying actors contributed to research and development capabilities and expected stable roles and predefined processes. In contrast, contractors adopted a more flexible approach, reallocating resources and shifting roles to meet project-specific needs. These differing business logics, process-driven versus project-driven, were associated with tensions, particularly in situations where reconfiguring the VP required actors to adapt quickly.

Process-industry capabilities were evident in the way one supplier operated an exclusive partner programme that provided participating contractors with business intelligence, pricing advantages, and prioritized access to resources. This arrangement also appeared to strengthen the supplier's position within the network.

For the main contractors, a central motivation for engaging with the shared VP was described as the ability to manage development risk through subcontractors by aligning contracts and integrating the network's specialized expertise into project delivery. This became particularly clear during discussions of a technically complex project, where such risk mitigation was viewed as essential for successful implementation.

> '. . ./In certain areas, we see the risk as too high, and then we must go with an outsourced assembly. And it's good to bring that in from the timber supplier so that we have the entire knowledge chain in place, rather than buying the assembly from one and the delivery from another, as parts can get lost in between.' (MC1)

However, the same customer and a partnered real estate developer were reluctant to allow new technical solutions to be tested in their projects, reflecting an expectation of risk aversion within the network. As one representative noted:

> '. . ./As a client, you're always a bit concerned and don't want to take too many risks, preferably no risks at all. Having methods tested in some way beforehand is a prerequisite for wanting to try something new.' (RE2)

This suggests that risk mitigation efforts continued to focus on contractual risk management, potentially at the expense of leveraging specialization and the network expertise as alternative ways to reduce uncertainty when introducing innovations.

4.2.3 Leadership capabilities

Ideally, the co-innovation network would support relationships not only among established actors but also with complementors who joined temporarily for specific building projects. Relational efforts targeting these complementors included both long-term initiatives and project-specific activities. Long-term activities were managed by the CLT-producers, which provided information and guidelines on their products to architects and technical consultants, aiming at getting involved, and their products specified correctly, in the early phases of building design, preceding the contract phase. In projects, collaboration between the focal firms and the customer's consultants was described as supporting the design process and contributing to efficient building projects.

In the early phase of one network, before a shared VP was in place, actors collaborated to strengthen competence and cooperation in the region, revealing the initial challenges they faced in building trust.

> '. . ./We [SE2, PW2, FC3] tried for example to initiate this with the regional university in some way. /. . ./, [PW1] was not really in existence at that time. [FC1] did not take part, but [CLT2] did. When we tried to engage these, that is about how I see this little network now. What I though think may be the most interesting thing to observe regarding the concept of co-operating firms is where you draw the line between /. . ./ competition and co-operation.' (PW1)

Later as the collaborations had evolved, trustworthiness was highlighted as crucial to business success by several of the respondents, some mentioning the importance to have good relationships over time when operating within a confined region with a limited amount of actors, while customers pointed out the trust of IE actors holding specialist competence in wood construction as critical for decisions on building with CLT.

The shared VPs were communicated to foster engagement and strengthen actors' commitment, while also clarifying the network's intended market position. One focal actor explained this as follows:

> '. . ./Then we have this slightly niched area where we offer assembled timber frames on ABT-U [Swedish standard contract for subcontracted design-build]. /. . ./ Where we have tried to attract expertise. So that we can, like, become the best at building in timber. And I would dare to claim that in our region, we have probably reached that point, I think. That we are at the forefront compared to other contractors, and that is probably also why we get these requests from other national contractors and other contractors.' (FC1)

How this communication was carried out and perceived by customers remained unclear. The data also pointed to inconsistencies, as several actors described the focal value proposition as unevenly communicated. Even in discussions concerning design–build contracts for CLT-based structural framings, one main contractor noted:

> '. . ./timber framing suppliers are today more or less just material suppliers. They do not have the holistic approach that we get from other types of framing systems.' (MC1)

Competition-related challenges emerged when marketing and negotiating the focal value proposition. At times, customer organizations, also positioned as contractors, viewed themselves as competitors to the focal actor, making early procurement discussions difficult. Some customers attempted to bypass the focal actor altogether, expecting CLT suppliers to deliver both assembly and technical services. To navigate this dilemma, the network occasionally supported less experienced contractors by offering assembly training and technical guidance through actors other than the focal firm.

4.3 Governance structure

Coordination and governance in the studied IE setting were observed to manifest through recurring arrangements, e.g., working routines, incentive negotiations, and information flows, that enabled or were associated with alignment of actors' contributions across overlapping projects. Governance appeared as repeated interactions rather than a centralized system and was described by respondents as yielding additional benefits to the actors compared with traditional governance structures for construction.

The IE:s VP was configured during the co-innovation activities within and in between projects and enacted through a project-specific reconfiguration by a subset of IE actors in individual building projects, starting with the procurement phase. More broadly, unlike traditional design-build approaches, where the VP is both configured and enacted in individual building projects and governance activities typically can be related into specific project phases, the observed IE governance mechanisms seem to be implemented iteratively within and across projects. These follow

TABLE 4 Conventional governance in construction industry versus IE-based governance.

Dimensions	Conventional governance	IE-based governance
Co-innovation	Executed by temporary organizations within individual projects to meet project-specific requirements	Executed by interdependent IE actors within, and between individual projects, motivated by actors' individual long-term goals as well as a shared goal of improving the shared VP
Learning	Ad-hoc, within projects, low knowledge transfer between projects	Structured knowledge transfer between actors and projects supported by the consistency in collaboration
Value proposition	Standardized components integrated into unique systemic VPs (buildings)	Standardized components integrated into a systemic VP
Contractual agreements	Project-based	Project based, supported by long-term contracts
Value capture	Distributed within building projects through contracts	Distributed within building projects through contracts, additional efficiency benefits from scaling principles and resource sharing realized through shared capabilities within the long-term collaboration

conventional project governance mechanisms, while continuously adapting for future projects: co-innovation is executed within building projects, but are further developed and standardized in between projects; learnings are transferred, and implemented as operational capabilities through the sustained collaboration to coming projects; the value proposition is standardized to fit a multitude of project settings but adapted to the unique projects; contracts are negotiated for project deliveries but often relying on fixed long-term contracts or agreements; and value is captured from the project delivery, but amplified through scaling principles and shared resources.

Table 4 summarizes the observed governance characteristics of the IEs compared to traditional governance structures across five dimensions that together outline the path from co-innovation to implementation: co-innovation, learning, value proposition, contractual agreements, and value capture.

5 Discussion

This section discusses the results of the thematic analysis, structured according to the dynamic capabilities framework. The analysis (Table 3) was consolidated into five themes for capabilities and subsequently into three levels of aggregated dimensions for DCs in CLT networks.

5.1 Operational capabilities for value creation

Indicative activities at this level included configuring the focal VP for project-specific solutions and coordinating resources to meet shared building project objectives. These practices reflect the ability to adapt offerings during procurement and design phases, consistent with Harty (2005) view of routines as embedded in project delivery. Such activities were triggered by client requirements and technical conditions, illustrating how operational routines evolve through situated negotiations. While these adaptations created possibilities for new business opportunities, they also introduced tensions around incentive alignment (Jacobides et al., 2024), echoing Ben Letaifa (2014) discussion on difficulties for emerging ecosystem actors to shift from traditional competition over value capture to coopetitive practices, which eventually may hinder the development of an IE.

The CLT-producers experienced reductions of volumes in their delivery scope following the IE's reconfiguration of the value proposition both during the procurement phase, and in the detail design phase. As contracts between the CLT-producer and the focal firm typically were volume-based, these optimizations also reduced profits accordingly. However, the CLT-producers acknowledged that these reductions were necessary to win the contracts. They also reported other efficiency benefits (i.e., reducing project management and sales costs) from the sustained collaboration, which appeared to compensate for the reduced volumes. However, these findings reflect the negotiations on value capture distribution at a given point of time within the lifecycle of the IE.

Overall, the focal actor and its suppliers pooled resources for co-innovation while developing the operational capabilities needed to embed these innovations into the VP.

5.2 Dynamic capabilities for developing operational capabilities

Observed practices at this level involved engaging in joint development efforts, implementing new technologies, and creating incentive structures to support innovation implementation. Examples include collaborative integration of moisture membranes into CLT production and co-development of prefabricated floor slabs, supporting Toppinen et al. (2022) findings on the perceived importance of prefabrication as an outcome from ecosystem collaborations in timber construction. These activities illustrate how actors leveraged network relations to extend individual VPs beyond single projects, aligning with Teece et al. (1997) notion of resource reconfiguration but enacted through informal coordination, distributed among actors, rather than formal central governance. However, the distribution of benefits was uneven, and complementors struggled to access joint development processes, highlighting the fragility of collaborative arrangements in temporary networks (Dubois and Gadde, 2002b). This suggests that while interactional enactments and activities may support innovation scaling, they require continuous relational work to sustain engagement.

5.3 Dynamic capabilities for strategic integration

Activities included facilitating knowledge exchange, aligning actor roles, and managing structures to enable co-innovation. Dissemination efforts through regional business promotion organizations and informal learning channels reflect attempts to build shared understanding, yet structured routines for knowledge transfer remained absent. Actor alignment was particularly challenging due to differing business logics, e.g., process-driven CLT suppliers versus project-driven contractors and the contractual reconfigurations in the main contractor's procurement process, creating tensions in role negotiation and risk management. Activities, roles, and positions within the network needed to be clearly defined and mutually recognized by participating actors. Leadership signals were weak and inconsistent, with early entrepreneurial initiatives failing to evolve into sustained strategic orchestration for co-innovation. This contrasts with IE literature that assumes a focal actor stabilizes the VP (Adner, 2016; Foss et al., 2023), suggesting that in construction IEs, co-innovation may remain vulnerable to fragmentation once initial innovations are implemented. This vulnerability appears related to the reliance on recurring but temporary project-based alignments rather than through a centralized or continuously stabilized structure. As actors shift roles between projects, shared routines depend on situated interactions within each project cycle, making collective capabilities vulnerable once an innovative effort comes to an end. Consequently, this opens for studying DCs for construction IEs as dynamically evolving configurations that cohere when projects overlap but may return to earlier forms of fragmentation when these linkages reduce.

5.4 Implications for CLT adoption

The findings indicate that co-innovation in timber construction does not primarily arise from centralized orchestration, such as policy-driven requirements for increased timber use. Instead, it appears to be driven by distributed and non-exclusive collaboration, where firms integrate learning across projects to strengthen their overall VPs. This supports Ilgın and Karjalainen (2024) argument that innovations are developed and operationalized within firms, while contrasting with traditional views of innovation coordination in construction (Harty, 2005) and IE perspectives that privilege focal actors as ecosystem orchestrators (Adner, 2016).

At the same time, the results show that such collaboration is context dependent. Capability development relies on interaction, trust, and cross-project learning. These conditions cannot be assumed to exist across regions or actor constellations. Scaling CLT without any form of coordination beyond individual projects may therefore be challenging, particularly in less matue IEs where interdependencies, risk and responsibility allocation remain contested.

Theoretically, this suggests that accelerating CLT adoption depends not solely on formal governance structures nor exclusively emergent, distributed co-innovation. Rather, it may require governance that combines capability development at the firm level with selective forms of coordination at the ecosystem or sector level. Such coordination does not imply centralized control, but involves shared standards, incentives, or arenas for interaction that reduce uncertainty and support alignment around the focal VP.

For managers, the findings suggest that scaling CLT is unlikely to be achieved through project-level delivery alone. Firms must invest in integrative (Helfat and Raubitschek, 2018) or interactive (Qiu et al., 2022) DCs that enable alignment, learning, and co-innovation across projects, while at the same time engage in structures that support collaboration over time. For actors seeking sectoral transition, the role may therefore be less about imposing control and more about enabling coordination mechanisms that reinforce co-innovation within evolving IEs.

5.5 Contributions

This study applies and develops a practice-based approach to studying DCs in construction industry, as suggested by Green et al. (2008), by showing how co-innovation in CLT-based construction emerges through situated, cross-project interactions such as value proposition reconfiguration. The notion of situated practices helps clarify that capability development is tied to recurring cross-project relations within the IE, rather than to centrally coordinated or formally institutionalized processes. This responds directly to calls for more contextualized and practice-oriented applications of DC theory (Schilke et al., 2018) in project-based industries (Robson et al., 2024). This practice-based contextualization of DCs constitutes a methodological contribution to innovation research in construction, offering deeper understanding of co-innovation and a promising tool for studying innovation implementation. This framework may also be applicable for investigating ecosystems within other subsectors of construction industry, since it addresses general construction industry innovation challenges (Hughes and Stehn, 2019) by suggesting the IE as a possible organizational alternative for overcoming these. The study complements single-project research on ecosystems in built environment (e.g., Toppinen et al., 2022) by offering an analytical framework capable of capturing how timber-related innovations are collaboratively developed, adapted, and scaled across temporary project settings. It advances theory at the intersection of DCs, IEs, and sustainable construction (e.g., Linde et al., 2021) by adopting the firm-level perspective on capabilities, which may help researchers observe and explain how firms operationalize sustainable construction practices. The findings indicate IE-level alignment mechanisms such as recurring collaboration, shared design practices, and continued interaction beyond single projects. These mechanisms may represent early indicators of processes that could support the accumulation of innovation-related effects over time, even though such dynamics are not evaluated in this study.

Managers operating in timber-based construction networks should focus on enabling practical coordination across projects by supporting shared routines, informal collaboration, and adaptive problem-solving. Clear communication and carefully designed incentive structures are essential for maintaining trust and managing differing perceptions of risk and responsibility among actors. Informal leadership distributed among actors, and trust-building should therefore be recognized as central alignment mechanisms rather than supplementary managerial activities. Public-sector actors can play an enabling role by reducing collaboration barriers and creating arenas for repeated interaction, particularly by investing in relational and informational infrastructure rather than imposing rigid coordination structures. These practices may create conditions that facilitate collaborative adaptation and learning, potentially supporting innovation beyond individual projects that allow timber-related innovations to scale across projects.

5.6 Limitations and further research

This study is subject to two main limitations. First, the analysis is based on a single case within a Swedish CLT-based construction IE, which makes the findings context-specific and interpretive rather than broadly generalizable. DCs are examined indirectly through observable practices and practitioner narratives, which provide insight into enacted co-innovation but cannot fully capture underlying DC structures. Although data triangulation through multiple interviews, researcher triangulation through independent analysis, and expert validation during the focus groups were employed, the findings should be viewed as indicative rather than definitive accounts of capability development. Second, the integration of DC theory with an IE perspective introduces conceptual challenges, particularly in distinguishing firm-level capability formation from ecosystem-level coordination mechanisms. This limitation reflects an inherent theoretical tension rather than a methodological weakness and points to the need for future research that more explicitly theorizes multilevel dynamics linking firm capabilities, co-innovation practices, and ecosystem governance in sustainable construction contexts. Further, as this study investigates co-innovation activities covering a fairly limited period of time, and does not include their temporal development, future research should explore the longitudinal development of enacted co-innovation processes and their effects on timber construction.

6 Conclusion

This study examined how DCs can support co-innovation within and across building projects in CLT-based construction. The findings show that co-innovation is enabled through situated practices operating at three interrelated capability levels: operational activities involving value proposition adaptation and resource coordination, mid-level practices of joint development and informal alignment, and strategic efforts focused on knowledge exchange and actor alignment across projects. Together, these practices demonstrate that DCs in construction can be traced within enacted processes through which co-innovation can be sustained beyond individual projects. Investigating DCs through situated practices provides a practical contribution to construction research, as it enables the capturing of co-innovation activities and routines within firms, ecosystems, and the temporary building project organizations.

The study further concludes that advancing sustainable timber construction and accelerating the adoption of EWPs may be achieved by interactive dynamic capabilities embedded within IEs, in addition to traditional approaches through formal governance structures or centralized orchestration. Pursuing sustainability ambitions requires attention not only to technological advances in EWPs, but also to the ecosystem-level conditions that may enable firms to learn, align, and co-innovate across temporary project settings in construction. By foregrounding these organizational and relational mechanisms, the study contributes a complementary perspective to the timber construction literature and highlights how co-innovation within IEs can support the broader transformation of the built environment toward sustainability.

Data availability statement

The datasets presented in this article are not readily available because the participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available. Requests to access the datasets should be directed to anna-lena.gull@associated.ltu.se.

Author contributions

A-LG: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review and editing. JE: Conceptualization, Formal Analysis, Methodology, Supervision, Validation, Writing – original draft, Writing – review and editing. LS: Conceptualization, Formal Analysis, Funding acquisition, Methodology, Supervision, Validation, Writing – original draft, Writing – review and editing.

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